

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EC) Examination

November 2013

EC404 Embedded Systems

Date: 19.11.2013, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

- Q - 1 (a) What is design metrics? Define four different design metrics of embedded systems. [03]
- (b) Define following terms: [04]
- i) Task Control Block, ii) Inter Process Communication
- Q - 2 (a) Explain Shared data Problem and its solution using Semaphore. [07]
- (b) Discuss the following content of I2C bus: [07]
- i) structure, ii) electrical interface, iii) format of an I2C data frame, iv) application interface.

OR

- Q - 2 (a) Explain Processor technology which used for designing Embedded Systems. Also compare all three major category named i) General Purpose, ii) Single Purpose, iii) Application Specific. [07]
- (b) Interface EEPROM 2708 with 8051 microcontroller. Also write a 8051 program to write and read the data from 2708 using I2C protocol. [07]
- Q - 3 (a) Write short note on CAN bus. [07]
- (b) What is RTOS? How does it differ from conventional OS? Describe three function and their activities provided by RTOS services. [07]

OR

- Q - 3 (a) With neat diagram, Interface LPC2148 with 4*4 Keypad, also write a C program to detect the pressed key. [07]
- (b) With the help of example, Explain following RTOS scheduling [07]
- i) Cooperative Scheduling of Ready Tasks using Precedence Constrains,
- ii) Preemptive scheduling.

SECTION – II

- Q - 4 (a) What are the main phases of a design review? [03]
(b) Briefly describe the differences between the waterfall and spiral development models. [04]
- Q - 5 (a) Interface LCD with LPC2148 using only 6 I/O pins. Write a program to display "CHARUSAT IS BEST". [06]
(b) Compare RISC and CISC processor. Why the RISC is more suitable for embedded system like network processor? [04]
(c) With help of suitable example explain all possible case in stack addressing. [04]

OR

- Q - 5 (a) What is advantage and disadvantage of pipeline? With neat sketch explain 3-stage pipeline and its drawback. [06]
(b) With help of suitable example explain data processing instruction of ARM processor. [04]
(c) Discuss the importance of "time-to-market" design metric with the help of example. [04]
- Q - 6 (a) Write a Technical note on Quality Assurance in Embedded System Design. [07]
(b) Differentiate general purpose processors and DSP processors. [04]
(c) Differentiate ARM mode and thumb mode. [03]
